

MACHINE LEARNING

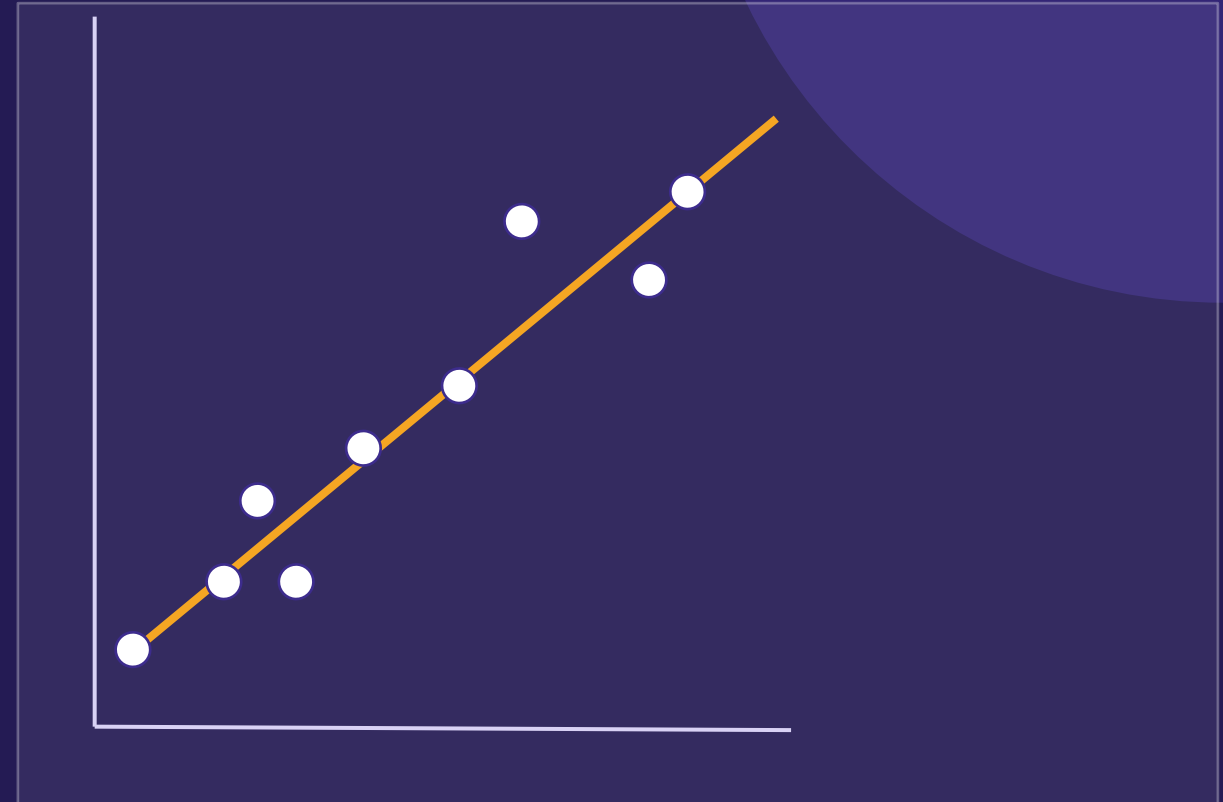
Linear Regression

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Data points • Best-fit prediction line

“Machine intelligence is the last invention that humanity will ever need to make.”

What Will We Learn?



1. What is Machine Learning?



2. What is Regression?



3. What is Linear Regression?



4. Understanding X and Y



5. The Best-Fit Line



6. Making a Prediction



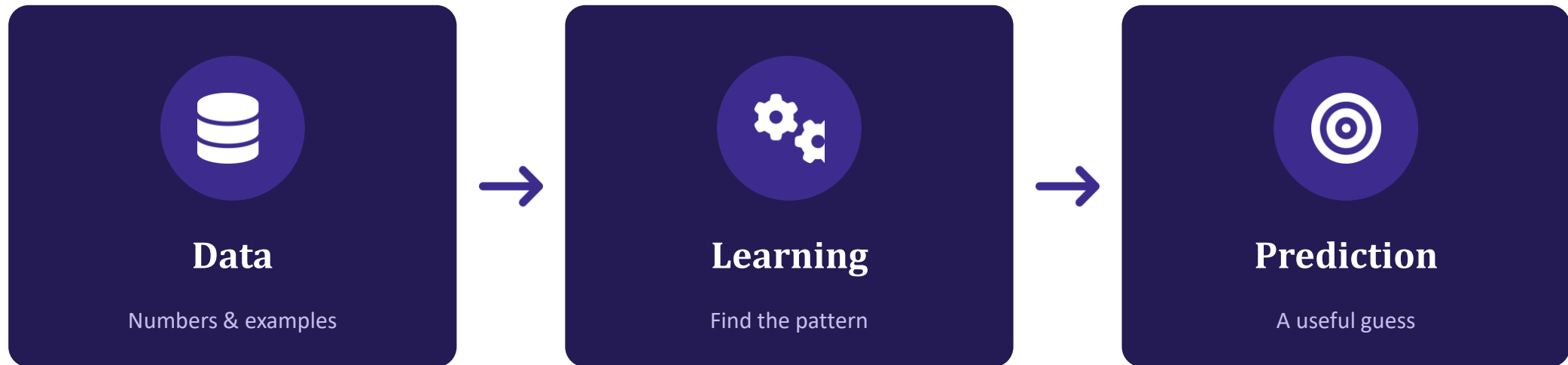
7. Python Implementation



8. Real-World Applications

What is Machine Learning?

Machine Learning allows computers to **learn patterns** from data and make **predictions**.



What is Regression?

Regression is used when we want to predict a **numerical value**.



House Price



Salary



Exam Marks



Temperature



Sales



Regression = Numbers

Predicts a value, like 62 marks or ₹35,000

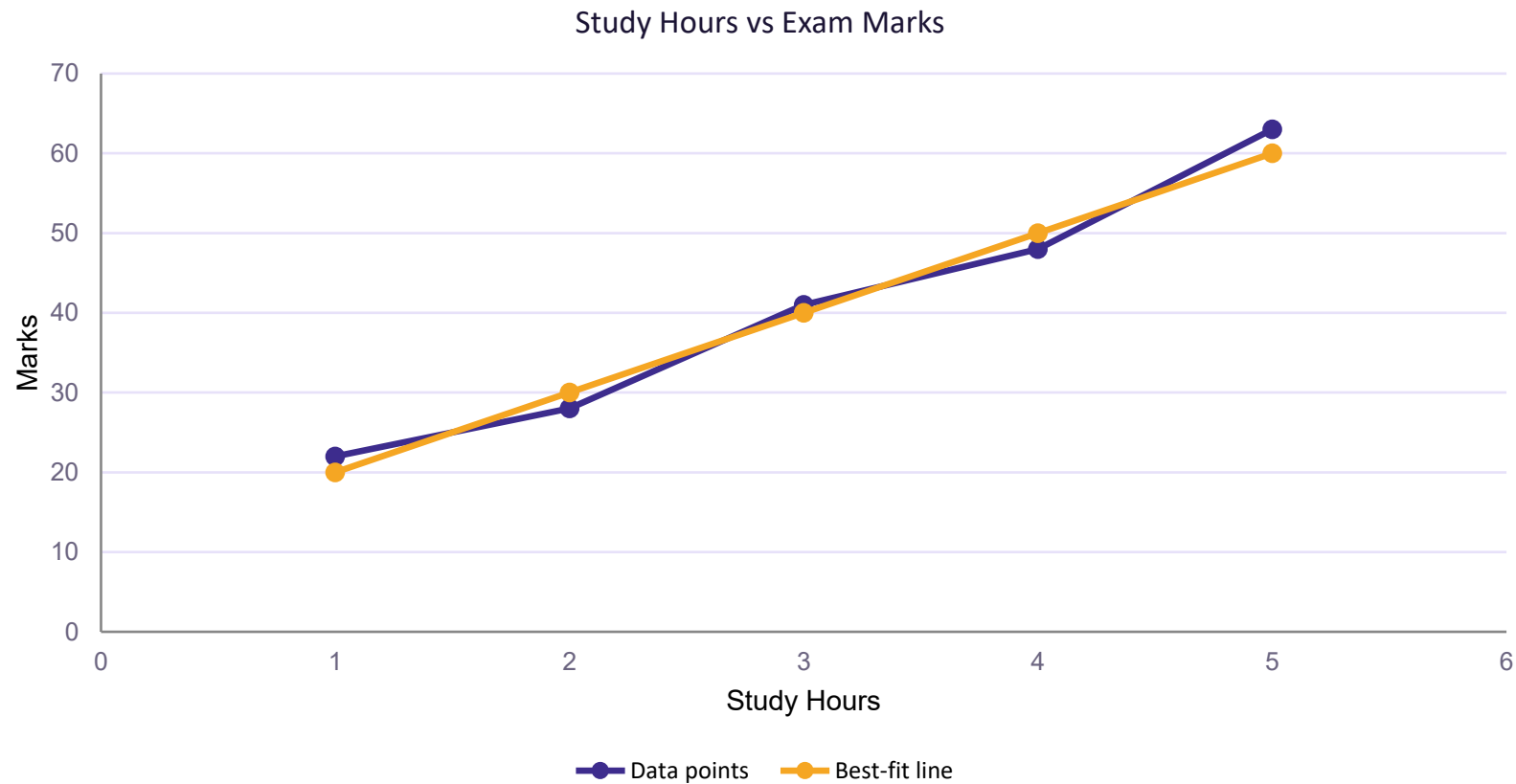


Classification = Categories

Predicts a label, like "Spam" or "Not Spam"

What is Linear Regression?

Linear Regression finds the relationship between input and output using a **straight line**.



Study Hours → Exam Marks

Study Hours	Marks
1	20
2	30
3	40
4	50
5	60



Notice the Pattern

Every extra hour of study adds about 10 more marks.

More study hours



Higher predicted marks

Understanding X and Y

X

Input / Feature

The information we already know

Example: Study Hours



Y

Output / Target

The value we want to predict

Example: Exam Marks

Understanding the Data Points

Each point on the graph represents **one real observation**.

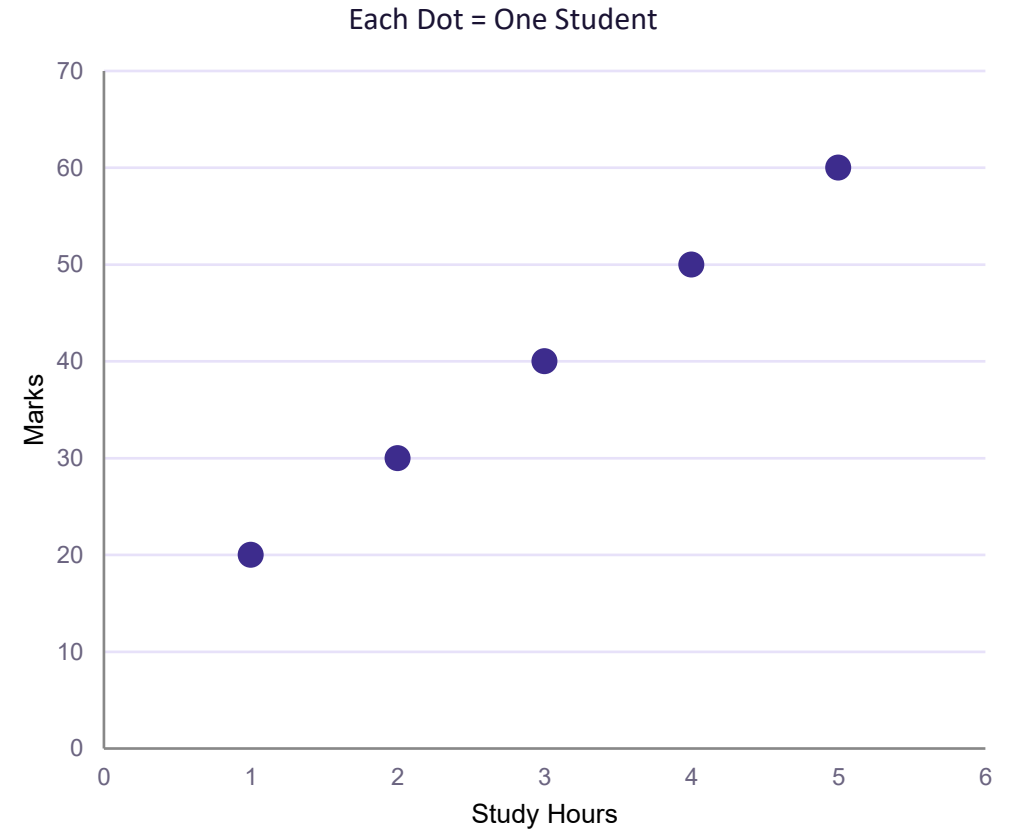
● (1, 20) 1 study hour → 20 marks

● (2, 30) 2 study hours → 30 marks

● (3, 40) 3 study hours → 40 marks

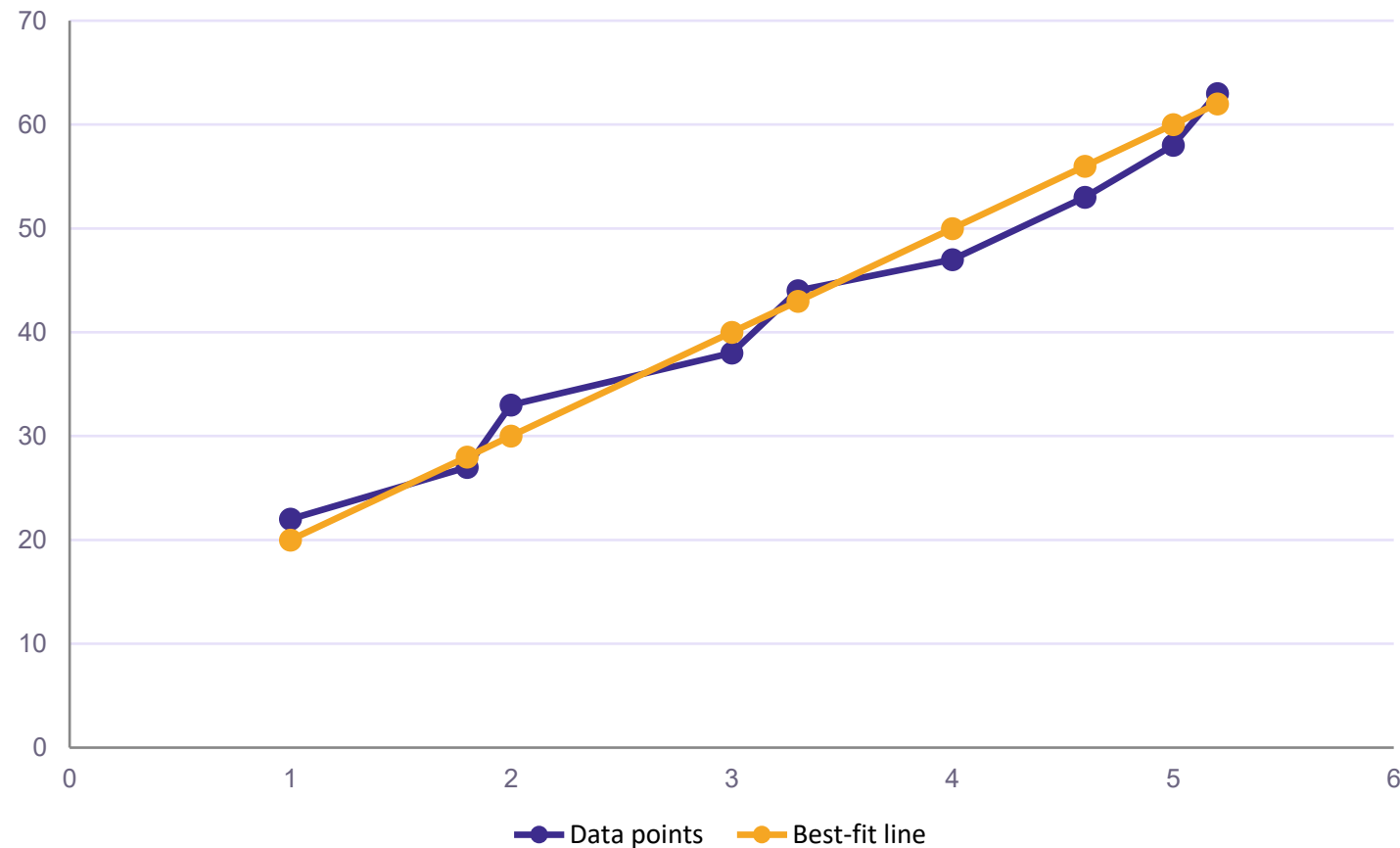
● (4, 50) 4 study hours → 50 marks

● (5, 60) 5 study hours → 60 marks



The Best-Fit Line

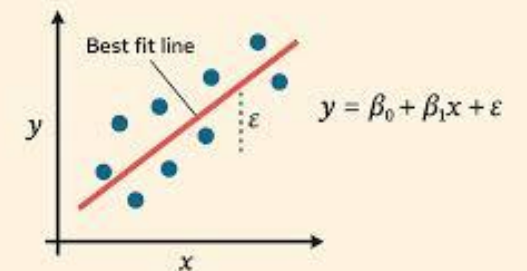
The best-fit line is the line that represents the **overall relationship** in the data.



Points close to the line → good fit

The line does not need to touch every point — it just needs to fit the trend.

LINEAR REGRESSION



Linear Regression Formula

$$\hat{y} = b_0 + b_1x$$



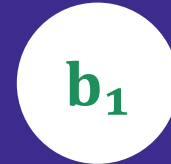
Predicted value

What we are guessing



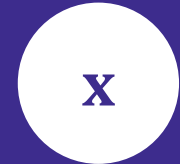
Intercept

Starting value, when $x = 0$



Slope

Effect of X on Y



Input

The value we know

Prediction Example

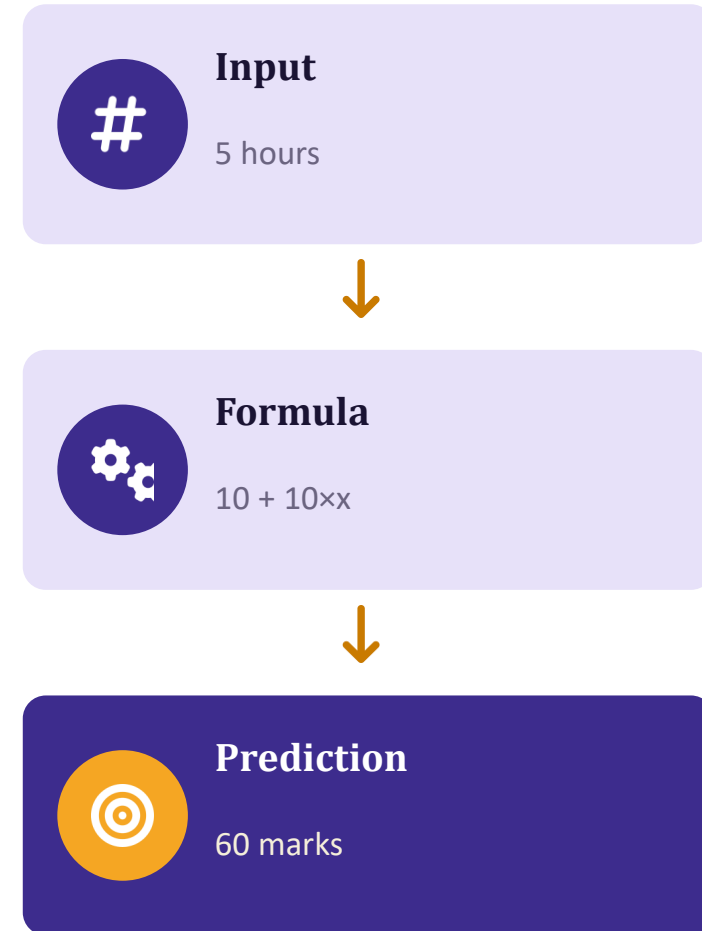
$$\text{Marks} = 10 + 10 \times \text{Study Hours}$$

For a student who studies 5 hours:

$$\text{Marks} = 10 + (10 \times 5)$$

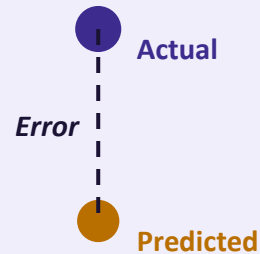
$$\text{Marks} = 10 + 50$$

$$\text{Marks} = 60$$



What is Error?

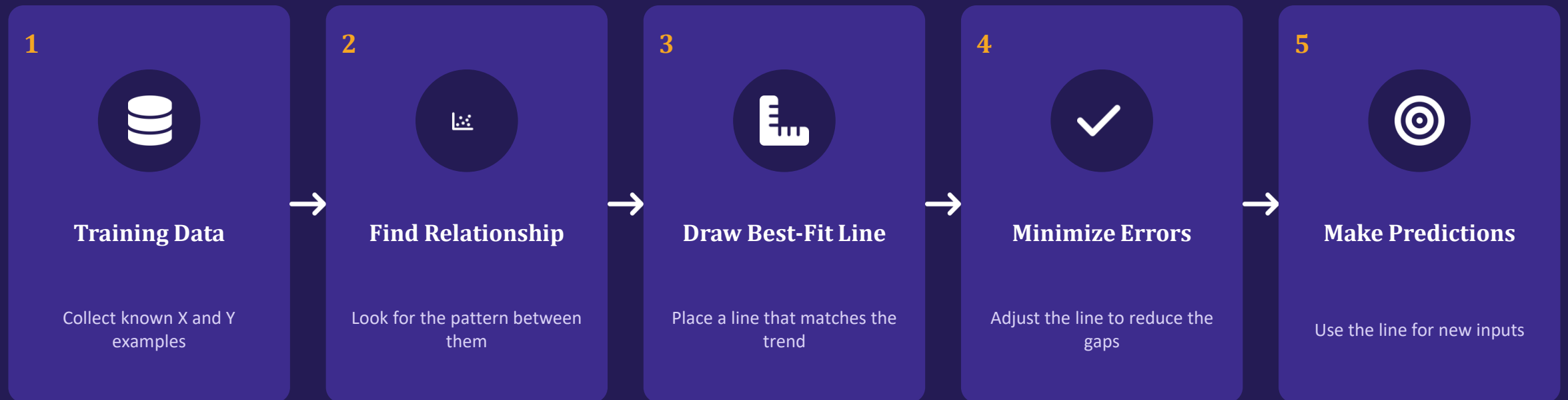
$$\text{Error} = \text{Actual Value} - \text{Predicted Value}$$



Why Error Matters

A good model keeps this gap as small as possible for every point. This gap is also called the "residual."

How Does the Model Learn?



Linear Regression in Python

We will use a ready-made library called **Scikit-learn** to do the maths for us.

```
from sklearn.linear_model import LinearRegression
```

```
X = [[1], [2], [3], [4], [5]]
```

```
y = [20, 30, 40, 50, 60]
```

```
model = LinearRegression()
```

```
model.fit(X, y)
```

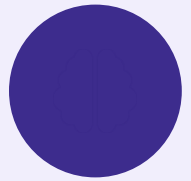
```
prediction = model.predict([[6]])
```

```
print(prediction)
```



X holds our study hours, y holds our marks — fit() teaches the model, predict() asks a question.

Understand `fit()` and `predict()`



`fit()`

Learn from data 🧠

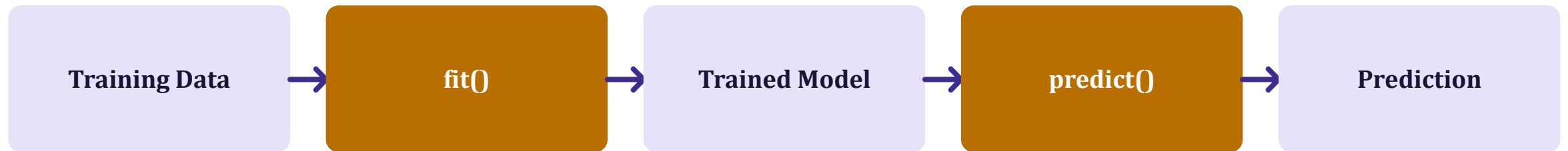
The model studies the examples and works out the best b_0 and b_1 values.



`predict()`

Make a prediction 🌟

The trained model uses its line to guess Y for a brand-new X.



Real-World Applications



House Price Prediction



Salary Prediction



Sales Forecasting



Car Price Prediction



Temperature Prediction



Business Forecasting

Beginner Activity



If a student studies 7 hours, what marks might the model predict?

Study Hours	Marks
1	20
2	30
3	40
4	50
5	60



Identify X

What is the input in this problem?



Identify Y

What is the output we are predicting?



Make a Prediction

Use $\text{Marks} = 10 + 10 \times \text{Hours}$ to solve it.

Quick Revision / Quiz

- 1 What is Regression?
- 2 What does X represent?
- 3 What does Y represent?
- 4 What does fit() do?
- 5 What does predict() do?



Quick Self-Check

If you can answer all five, you understand the basics of Linear Regression!

Linear Regression = Learn the relationship → Draw the best-fit line → Predict a number.



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